Slave-boson convergence audit

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Outcome: the numerical checks pass within the declared mean-field equations. The audit supports converged saddle solutions and selected local stability; it does not bound their error relative to the exact interacting impurity.

1. Finite-U matrix KR saddle

All 98 accepted stored records were re-solved at integration orders 32 and 64, then checked at order 96 without further optimization. This includes the low-filling and occupied paths to U=0.2, and both filling-one EP families. The count includes overlapping records in original and extended paths.

Check	Measured result
Largest elementwise change in R, order 32 → 64	5.14e-13
Largest change in controls / observables	1.38e-11 in model units
Largest normalized residual at order 96	5.14e-12
Tail cutoff 100,000 → 1,000,000	Density < 2.3e-16; force < 4.4e-18
Analytic radial integral / adaptive quadrature	1.54e-16
194-dimensional Hermitian star: density / force	1.14e-12 / 1.48e-14
Independent log-determinant derivative check	Largest discrepancy 5.07e-10
Initial seeds and local stability	Six perturbed seeds agree; tested constrained and transverse curvatures positive

Filling-one example: U=0.0025, n_d=1, double occupancy D=0.22894565, single-occupancy probability 0.54210869, $Z_{\pm}$=0.99739550. The pole is 0.0065351343 − 0.0029830358i. This is a charge-fluctuating EP, not a local-moment demonstration.

The separately tested occupied U=0.2 point has n_d=0.994055 and D=0.001260, but is strongly polarized and has split tested poles. Positive local curvature does not establish a unique global minimum.

2. Corrected scalar Coleman benchmark

This is the separate U→∞, Q=1 constant-core model: $\varepsilon_d=-1$, $\Gamma_0=0.12$, band cutoff D=4, $T=10^{-6}$. The archived convention is $\Delta_0=0.075+0.05\beta$ and $\Gamma_{PT}=0.2\beta$. Fixed $\Delta_0=0.10$ is also checked and separately labeled in the JSON.

Correction at β=0.5: r=0.00820293449, with $\delta_c-1=8.74563e-10$. The previous r=0.00980741173 overestimated the r^2-dependent scale by 42.95%. Its stationarity residual on the corrected grid is about 0.0264.

The old grid resolved narrow poles but missed the logarithmic occupied-band tail. The repaired grid refines that tail and the uniform base grid; the constraint root tolerance is tightened and residual acceptance is explicit.

β	Independent r	Finest-grid r	Relative error
0.35	0.00875360325	0.00875361098	8.83e-07
0.50	0.00820293449	0.00820294148	8.52e-07
0.55	0.00795604253	0.00795604929	8.49e-07

Three grid levels range from roughly 22,000 to 134,000 points. The finest solutions agree with the independent finite-cutoff reference to better than one part per million in r; independent residuals are below 1.4e-7. All three safeguarded production solves converge.

Finite-T digamma expressions are checked with the lower-edge correction −log(1+z/D), and independently against adaptive real-axis quadrature (saddle residual < 5e-15). The smooth low-T limit gives r(0)=0.008298290454. Passivity and the FDR/charge identities hold for the stated thermal Markov closure; they are consistency checks of that closure.

3. Scalar saddle of the full stationary bath

For effective W=0.1: $n_f=0.03468136$, r=0.98250630, $\delta_c=0.0006740880$, bare $\varepsilon_d=0.01449626$ and bare W=0.10178052. The local control-map determinant is 1.12916.

At fixed bare W=0.1, the retuned EP is $0.01177011658 - 0.00270876327i$, with joint residual 1.31e-14. Integration orders 48 and 96 differ by < 8.4e-15. Matsubara sums at T=0.001, 0.01 and 0.1, using 32k/64k/128k terms and analytic tails, change by at most 2.3e-12 from 64k to 128k.

The full-bath scalar r, the constant-core scalar r and the finite-U matrix R refer to different approximations and parameter sets.

4. Pole tracking and scope

Four EP loops were repeated with 256 intervals and integration order 64, in addition to the 128-sample calculations. The saddle is re-solved at each control point; frequency derivatives hold that saddle fixed.

Point / control	Degree	Half-winding	Permutation
Low filling, U=0.02; (εd, VO)	+1	+1/2	(12)
Low filling, U=0.2; (εd, VO)	+1	+1/2	(12)
Low filling, U=0.0025; (εd, W)	−1	−1/2	(12)
n = 1, U=0.0025; (εd, VO)	−1	−1/2	(12)

All four loops retain two open channels. The minimum $|F_{zz}|$ exceeds 1.72, pole residuals are below 5.4e-21, normalized saddle residuals below 6.0e-12, and the largest assignment/half-separation ratio is 0.177. Center kernels have rank one within the numerical tolerance.

SciPy lack-of-progress flags occur at 5 and 9 accepted loop solves for U=0.02 and 0.2. These flags remain in the output: acceptance uses the explicit residual, rank and tracking checks. The certificates are sampled numerical diagnostics, not interval proofs over every control value.

The fixed-W, exactly-filling-one fold is reproduced at U≈0.0030826663. A nearby charge-fluctuating EP at U=0.0032 remains present after releasing the exact filling constraint. The fold therefore does not prove global EP disappearance.

Finite-chain comparison: four N=12-mesh KR points are freshly recomputed. The ED centers used for comparison are stored reference inputs. Checking the historical N=10→12 contour summary is not a fresh ED continuation, and is not continuum convergence.

5. Reproduction and publication use

The ZIP includes all scripts and data for these mean-field checks, the corrected manuscript snapshot, pinned numerical dependencies, and an entry point: **python reproduce.py --verify-stored**, **--quick** or **--full**. Full details and individual outputs are indexed by VALIDATION_REPORT.md. The previous complete ED/NRG archive remains separately citable at DOI 10.5281/zenodo.21888925.

Suggested statement: **“The reported slave-boson EPs and saddle observables are numerically converged within the specified mean-field approximations.”** Exact local-moment EP survival, thermodynamic NRG survival and pumping-enhanced screening remain unestablished.

The manuscript and Supplement remain the corrected 23 October snapshot. Check the author affiliation and enter the actual new fixed-version DOI before journal upload. This candidate has not been pushed, deposited or submitted.